Geospatial Analysis of COVID-19 Vaccination, A Neighbourhood Level Analysis in Ireland

A Preprint

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Abstract

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1 Introduction and Background

COVID-19 vaccination has been one of the most impactful public health programs that helped mitigate the global burden of the pandemic. According to estimates published in Lancet Infectious Diseases, vaccines prevented approximately 20 million deaths in their first year of rollout, highlighting their role in reducing transmission, hospitalisation, and mortality in high-coverage settings (Watson et al. 2022). However, significant disparities in vaccine uptake have persisted across regions and population groups, reflecting deeper socio-political and structural divides.

The global COVID-19 vaccination campaign, while achieving remarkable population coverage, has revealed significant geographic disparities in uptake rates that cannot be explained by individual-level factors alone. This spatial variation in vaccination behaviour has become an important area of research, with implications for understanding health inequities and informing future public health interventions.

Spatial Patterns: Evidence from England (Dropkin 2022): Research examining spatial patterns in COVID-19 vaccination uptake has provided valuable insights into geographic disparities. A notable study of COVID-19 booster uptake across England found substantial variation, with Third Injection uptake rates ranging from 27.2% in Newham to 67.4% in Gloucestershire—representing a 40 percentage point difference between areas. This research examined area-level predictors beyond typical individual characteristics such as age, gender, ethnicity, and deprivation levels. The findings revealed that while individual socio-demographic factors remained significant predictors of vaccination uptake, considerable unexplained area-level variation persisted, suggesting that geographic factors play an important role in vaccination decision-making processes.

Research employing geographically weighted regression (MGWR) models (Chen et al. 2023) investigated how relationships between socio-demographic variables and vaccination rates vary across different geographic locations. The study found that socio-economic-demographic variables explained 83.2% of the variance in vaccination uptake patterns, with the strongest positive associations observed for proportion of population over 40 years, car ownership rates, and average household income. Importantly, this research demonstrated that the strength of associations between predictors and vaccination uptake varied significantly across geographic areas, indicating that traditional global statistical models may not adequately capture local variations in these relationships.

Barriers to Vaccination Access: Literature has identified that structural barriers, particularly geographic access, represent significant factors influencing vaccine uptake. Spatial and logistical constraints including distance to vaccination centres, limited public transport options, and regional disparities in health infrastructure have been identified as major obstacles to vaccination, even in high-income countries. Cascini et al. (2023)[5] conducted a systematic review which found that access-related challenges were particularly pronounced in rural and socioeconomically deprived areas. Similarly, Molteni et al. (2024)[6] demonstrated that physical proximity to vaccine centres remained an independent predictor of uptake in European urban and peri-urban zones.

International research has consistently documented spatial disparities in vaccination uptake across different countries and contexts. For example, de Figueiredo et al. (2021) used multilevel regression and post-stratification techniques to forecast vaccine hesitancy across the UK, successfully identifying clusters of low vaccination intent that correlated with ethnicity, economic insecurity, and urban location. In the French context, Lourenço et al. (2022)[8] found that vaccination uptake was significantly lower in regions characterized by higher poverty levels and greater migrant population density. These findings have led researchers to advocate for more granular, sub-county and neighbourhood-level modeling approaches to support more equitable public health responses. Nash et al. (2025)[9] specifically recommended incorporating travel-time variables and census-linked deprivation scores to identify vaccination "cold spots" that require targeted interventions.

Research Gap - There appears to be a significant research gap regarding spatial accessibility to vaccination sites and geographic disparities in vaccination uptake within the Irish context. This gap is particularly noteworthy given Ireland's unique geographic and demographic characteristics. The country's settlement pattern, which combines major urban centres (particularly the Greater Dublin Area) with rural and remote communities, creates distinct spatial challenges that may differ considerably from those documented in England or other international contexts. National statistics indicate that Ireland achieved over 90% uptake of at least one COVID-19 vaccine dose by the end of 2021, according to reporting by the Health Protection Surveillance Centre (HPSC)("2021 - Health Protection Surveillance Centre" n.d.). However, when data was examined at the more granular LEA level, spatial inconsistencies in coverage became apparent, particularly in areas characterized by higher deprivation levels and weaker healthcare access. While researchers have applied spatial analytical approaches to study COVID-19 infection patterns in Ireland, including spatio-temporal cluster analysis, the specific application of spatial modeling techniques to vaccination services remains largely unexplored.

This background establishes that spatial variation in vaccination uptake is a well-documented phenomenon internationally, with clear evidence of structural access barriers affecting uptake patterns. However, the Irish context remains underexplored, presenting an opportunity to contribute to the understanding of geographic factors influencing vaccination accessibility and uptake in a unique geographic and demographic setting. Our work responds to those calls by conducting a spatial and demographic analysis of COVID-19 vaccine uptake in Ireland, using LEAs as the unit of observation. By merging CSO-derived demographic indicators, geospatial accessibility scores, and HPSC vaccine coverage data, the study aims to identify patterns of inequity, assess their correlates, and support data-informed, geographically targeted public health strategies.

2 Methods

2.1 Study Design

This study employs a quantitative, observational, and ecological research design to explore variation in COVID-19 vaccination uptake across Ireland. The ecological framework is suited to the research aims, as the analysis is conducted at the population level rather than the individual level, with the LEA serving as the unit of observation. This design enables the investigation of associations between area-level characteristics—such as age, education attainment, gender and deprivation scores—and aggregated vaccination rates as outcomes.

Observational designs are appropriate when experimental manipulation is neither ethical nor feasible, as is the case with vaccine uptake during a pandemic. The study does not seek to infer causality, but rather to identify statistical associations and spatial patterns that may inform future public health strategies. The ecological perspective, while limited in its ability to capture individual-level behavioural determinants, provides valuable insights into structural and contextual influences that may affect access, acceptance, or hesitancy at the community scale.

Additionally, the study incorporates a temporal dimension by analysing vaccination uptake over successive months, enabling the identification of evolving patterns and time-based disparities in vaccine coverage. This temporally extended approach complements the spatial analysis and enhances the explanatory power of the models. The combination of spatial and temporal design elements positions this research within the growing body of data-driven public health studies aimed at understanding complex, dynamic health phenomena within real-world settings.

2.1.1 Data Sources

To address the research objectives, this study integrates multiple publicly available datasets capturing vaccination services, uptake rates, and geographic boundaries. The vaccination services data encompass three distinct service delivery models that formed Ireland's COVID-19 vaccination network during the rollout.

Initial Vaccination Centers for COVID-19: 35 large-scale vaccination centers established specifically for the national COVID-19 immunization program were identified and catalogued. These centers represented the primary delivery mechanism during the initial vaccine rollout phases, particularly for priority populations and mass vaccination campaigns.

General Practitioners (GPs): 1557 general practice clinics (GPs) participating in the COVID-19 vaccination program were systematically identified. GP-administered vaccines represented a crucial component of community-based delivery, particularly for reaching populations with established healthcare relationships and those requiring more personalized care approaches.

Pharmacies: 1041 community pharmacies authorized to administer COVID-19 vaccines were documented as part of the expanded vaccination network. Pharmacy-based vaccination services enhanced accessibility by leveraging existing community health infrastructure and extended service hours.

2.1.2 Data Collection Methods

All vaccination center data were obtained through systematic web scraping techniques implemented on the HSE website using rvest in R and Selenium in Python.

Static Web Pages: The rvest package in R was used for scraping data from static HTML pages. For static web pages, the information is directly available in the HTML source when the page loads, making rvest ideal for quick and simple extraction of data from the page's structure.

Dynamic Web Pages: Selenium in Python was utilized to handle JavaScript-rendered content, enabling access to dynamically loaded data not accessible via static scraping. Some websites use dynamic content, where information is loaded separately through JavaScript after the initial page is displayed. Selenium can simulate a real web browser, allowing the script to wait for content to load, click buttons, or scroll down the page to access hidden or delayed information. While rvest is faster and easier for static content, Selenium is necessary for interacting with and extracting data from more complex, dynamic sites.

All identified vaccination centers were geocoded to obtain their geographic coordinates (longitude and latitude). The Google Maps API was employed to ensure accurate and standardized geolocation across the dataset. A custom function was written to send HTTP requests to the Google API with each address and extract the corresponding latitude and longitude from the returned JSON data. This approach allowed precise control over the geocoding process and can be combined with rate limiting to comply with API usage policies.

Vaccination uptake data which is the LEA vaccination rates were sourced from table CDC47 published on the Central Statistics Office (CSO) website. This dataset provides vaccination coverage statistics at the Local Electoral Area level, representing the most granular administrative geography for which reliable vaccination data were consistently available throughout the vaccination rollout period.

LEA boundary files were obtained from Ordnance Survey Ireland (OSI), providing the official digital geographic boundaries necessary for spatial analysis and mapping applications. These boundary files serve as the foundational geographic framework for linking vaccination rates with corresponding service locations.

In parallel, socio-demographic variables—such as education, age, gender proportions and deprivation indices by LEA—were extracted from the CSO's 2016 Census and supplementary 2022 updates where applicable.

2.1.3 Engineering Accessibility Data

Traditional accessibility measures include naïve distance-based metrics and provider-to-population ratios, Luo and Qi (Luo and Qi 2009)established the modern spatial accessibility analysis method of two-step floating catchment area (2SFCA). They demonstrated that the 2SFCA method addresses critical limitations of traditional distance-based measures by incorporating both healthcare service supply (provider capacity) and it's demand (population need) factors simultaneously. This approach provides more accurate representations of healthcare accessibility patterns by accounting for competition among populations for limited healthcare resources. Their method first involves identifying all population locations within a reasonable travel distance of each healthcare provider, then calculation of provider-to-population ratios. In the second step, it identifies all providers within reasonable distance of each population location and sums their accessibility ratios. This dual consideration of supply and demand makes the 2SFCA method relevant for our study where both vaccination site capacity and population demand vary spatially.

Luo and Wang(Luo and Qi 2009)provided a comprehensive comparison of gravity models, enhanced 2SFCA methods, and other spatial accessibility approaches. Through their analysis in the Chicago region, it was demonstrated that different accessibility measures can produce substantially different results depending on geographic context, population distribution, and healthcare service characteristics. The study revealed that gravity models, which weight healthcare opportunities by distance decay functions, often provide more nuanced accessibility scores than simple measures. Wang's framework offered crucial direction for vaccination accessibility research by establishing important factors to take into account when choosing suitable spatial accessibility models depending on study goals and data availability.

Nevertheless, detailed demographic statistics at the sub-LEA level and, ideally, information regarding service capacity are necessary for the complete implementation of 2SFCA, and these were not available for this study. Furthermore, the approach is resource-intensive, requiring substantial geospatial processing and catchment dynamics assumptions. So, we decided not to use 2SFCA in our analysis as a result.

We used an intuitive spatial coverage measure based on proximity - travel isochrones were generated for each service location. Specifically, 10, 20, 30, and 60-minute drive-time isochrones were created for the initial vaccination center, while 10-minute drive-time isochrones were generated for pharmacies and GPs. Each LEA polygon was then intersected with these isochrones to determine the extent of overlap, representing the proportion of the LEA covered by the service catchment areas. The total overlapping area was measured to produce a naive accessibility score for each LEA. For the initial vaccination center, a more refined accessibility score was calculated by inversely weighting the overlapping areas by travel time, reflecting the assumption that closer proximity implies better access. This approach was based on the intuitive premise that any spatial overlap between a healthcare service's catchment and an LEA indicates at least some degree of physical access for residents.

Applying 2SFCA—or an advanced variation—would yield a more comprehensive assessment of healthcare accessibility with more specific data and specialized resources, particularly in contrasting rural and urban situations where demand and service burden varied significantly and we strongly advise using it in subsequent research.

2.1.4 Data Merging

The process of merging datasets from multiple sources constituted a pivotal step in preparing the analytical dataset used in this study. Each dataset—vaccination data, demographic variables, and accessibility scores—originated from distinct sources and differed in structure, temporal resolution, and formatting. To ensure analytical coherence, extensive preprocessing was undertaken to harmonise identifiers, align time frames, and integrate relevant variables into a single, unified data table.

The vaccination dataset, which served had the primary outcome data contained multiple records per LEA, each corresponding to a specific reporting month. This long-format structure was transformed to suit the objectives of the analysis. For our study we only focussed on the vaccination data for people 12 years and older and our primary outcome of interest was primary vaccine dose's vaccination rate.The socio-demographic variables were available in wide format, with a single observation per LEA.

The critical challenge in data integration arose from inconsistencies in LEA naming conventions across the vaccination uptake data (CDC47) and the geographic boundary files (OSI). To address this systematic issue, LEA names from both datasets were pre-processed to ensure consistency in naming conventions. This preprocessing workflow addressed variations in spelling, capitalization, punctuation, and formatting that would otherwise prevent successful dataset alignment. Following this standardization process, the two datasets were joined using the merge() function in R, with the harmonized LEA names serving as the common linking key to establish precise one-to-one correspondence between vaccination statistics and geographic boundaries.

The accessibility variables and sociodemographic variables were stored in separate datasets and were appended via the same standardized LEA identifier matching protocol used for the primary data integration. This resulted in two final datasets – one which contained one row per LEA, with columns for monthly rates of vaccination uptake for primary, booster 1, booster 2, booster 3 doses and LEA's geometric polygons. The second dataset contained all demographic variables, and the engineered accessibility score per LEA. These served as the analytical foundation for exploratory and modelling components of the study.

2.1.5 Model

The beta regression model was chosen to handle the bounded, proportional nature of Primary Vaccination Rate. In beta regression, the mean parameter models the expected value of the outcome on a logit scale (or other link functions), while the precision parameter accounts for how tightly values are clustered around the mean. Its logit link ensures predictions stay within [0,1], while the precision parameter accounts for heteroskedasticity. In zero-inflated beta models, an additional parameter, zi (zero-inflation probability), models the likelihood of structural zeros (i.e., excess observations at exactly 0 or 1). As our data did not contain exact zeros or ones, we did not use this.

Demographic factors were compositional proportions, so they were additive log ratio transformed to enable their inclusion in the models. The baseline chosen for the transformation were the following: - Age: 71+ year olds - Education: Post Graduate Level - Health Outcome: Very Good - Sex: Female

In the initial beta regression model, we examined the relationship of primary vaccination rate with age and sex. Then it was adjusted with predictors of health, education level and accessibility to initial vaccination center. Both utilized Markov Chain Monte Carlo (MCMC) sampling with 4 chains of 2000 iterations each, including 1000 warmup iterations, yielding 4000 post-warmup draws for posterior inference.

Morans'I was calculated to identify if spatial autocorrelation existed in the residuals. The presence of spatial autocorrelation led to testing spatial models, specifically Conditional Autoregressive (CAR) ("Spatial Conditional Autoregressive (CAR) Structures — Car" n.d.)priors, which allow neighboring LEAs to share information through structured random effects based on an adjacency matrix of LEAs. This model utilized extended MCMC sampling with 30,000 iterations per chain and 10,000 warmup iterations, yielding 80,000 post-warmup draws to ensure adequate exploration of the posterior distribution for the more complex spatial model. We examined the role of socioeconomic deprivation beyond individual education measures, by extending the non-spatial model by incorporating a composite deprivation index .

We also utilized the joint distribution of age and sex as predictor variables in addition to examining these factors in isolation. For instance, vaccination attitudes among young men may differ substantially from those of older women, and analyzing age and sex as separate variables would fail to detect these important interaction patterns. To examine potential differential effects of age across gender groups, we fitted a model using sex-specific age categories rather than separate age and sex terms. This approach allowed for the identification of age patterns that may vary between males and females, moving beyond the additive assumption of previous models.

We incorporated joint age-sex proportions (such as women aged 55-64 or men aged 18-24) to better represent population heterogeneity and its relationship with vaccination behaviour. They were proportions and were also additive log-ratio transformed to make them suitable for regression analysis.

We captured the nonlinear trajectory of primary COVID-19 vaccination uptake over time and across different regions, by specifying a nonlinear growth model using a sigmoidal function of time (month-level) fitted within a hierarchical Bayesian framework. The outcome variable was modeled using a beta regression with a four-parameter logistic function of the form:

Primary_Vax_Rate ~ base + (Asym - base)/(1 + exp((xmid - Month_num)/scal))

where base represents the lower asymptote (initial vaccination rate), Asym represents the upper asymptote (maximum vaccination rate), xmid denotes the inflection point (i.e., the month at which uptake growth is steepest), and scal controls the slope or spread of the curve.

To account for spatial variation and demographic influences, the model incorporated hierarchical structure with group-level random intercepts for each LEA. Asym was modeled as a function of demographic covariates including ALR transformed age distribution ratios and male population ratio, with LEA-specific random intercepts. Both the base and xmid included LEA-specific random effects to capture regional variation in initial uptake rates and timing of vaccination acceleration. The scal was estimated globally across all regions.

The analysis employed longitudinal data with 4,980 observations across 166 LEAs over multiple time points and the model was fitted using No-U-Turn Sampler (NUTS) with 4 chains, each with 4,000 iterations and 2,000 warmup iterations, yielding 8,000 post-warmup draws.

The models were all implemented in brms (Bayesian Regression Models using Stan) (Heiss n.d.; "Parameterization of Response Distributions in Brms" n.d.) package in R as it offers a flexible Bayesian framework ideal for modeling complex structures like spatial, temporal, and hierarchical effects; it supports beta regression for proportion outcomes and can easily extend to other model families, making it well-suited for our geographically structured vaccination data.

3 Results

3.0.1 Exploratory data analysis

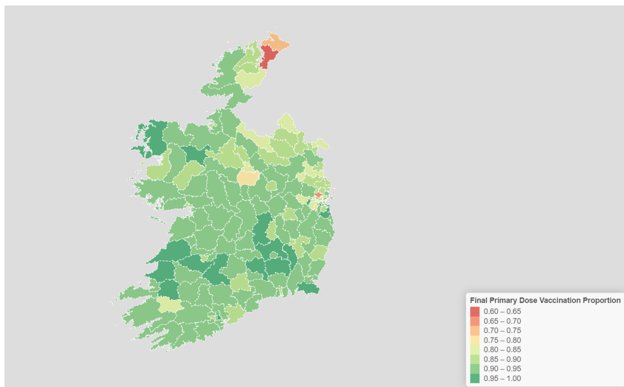

Figure 1: LEA wise primary dose final vaccination proportion

3.0.2 Model

In the initial beta regression model, we examined the relationship of primary vaccination rate with age and sex. Younger age groups demonstrated significantly lower vaccination rates compared to the reference category, with particularly strong negative effects for the 18-54 year age group (-1.42 log-odds, 95% CI: -1.75 to -1.07) and the 12-17 year age group (-0.60 log-odds, 95% CI: -1.00 to -0.24). The 55-64 year olds exhibited substantially higher vaccination rates (2.48 log-odds, 95% CI: 1.49 to 3.45), a strong positive predictor in the model. The 65-70 year olds showed a modest negative coefficient (-0.42 log-odds, 95% CI: -1.11 to 0.30), though the credible interval (CI) included zero, suggesting uncertainty around this effect. Male proportion demonstrated a negative association with vaccination rates (-1.77 log-odds, 95% CI: -3.52 to -0.08), with the CI marginally excluding zero.

In the model adjusted with health, education level and accessibility to initial vaccination center, we could see that the age effects persist but attenuated slightly while revealing additional significant associations. Upper secondary education emerged as a significant positive predictor (1.34 log-odds, 95% CI: 0.53 to 2.16), suggesting higher vaccination rates among populations with this educational level. Conversely, areas with higher proportions of individuals with no formal education showed lower vaccination rates (-0.47 log-odds, 95% CI: -0.83 to -0.11). Health status variables showed no statistically significant associations with vaccination rates. While very bad and bad health were positively associated, and fair/good health negatively associated, all credible intervals included zero, indicating high uncertainty and weak effects.The male proportion effect changed dramatically between models, shifting from negative in the initial model to positive (1.39 log-odds, 95% CI: -0.73 to 3.45) in the adjusted model, though with substantial uncertainty as evidenced by the wide confidence interval. Accessibility to initial vaccination centers showed negligible impact (-0.03 log-odds, 95% CI: -0.09 to 0.03), with the credible interval centered on zero, suggesting that geographic accessibility was not a significant barrier to vaccination uptake in our population. The precision parameter (phi) increased from 57.18 (95% CI: 45.23-70.46) in the initial model to 77.74 (95% CI: 61.18-97.02) in the expanded model, indicating improved model fit and reduced residual variance.

Mapping the residuals from the adjusted beta regression model revealed clear spatial clustering—neighboring LEAs often shared similar under- or over-predictions. To formally test for spatial autocorrelation, we conducted a Moran's I test on the residuals: Moran's I = 0.1906, $p < 0.001$ — indicating significant positive spatial autocorrelation in residuals (i.e., LEAs geographically close to one another tend to have similar

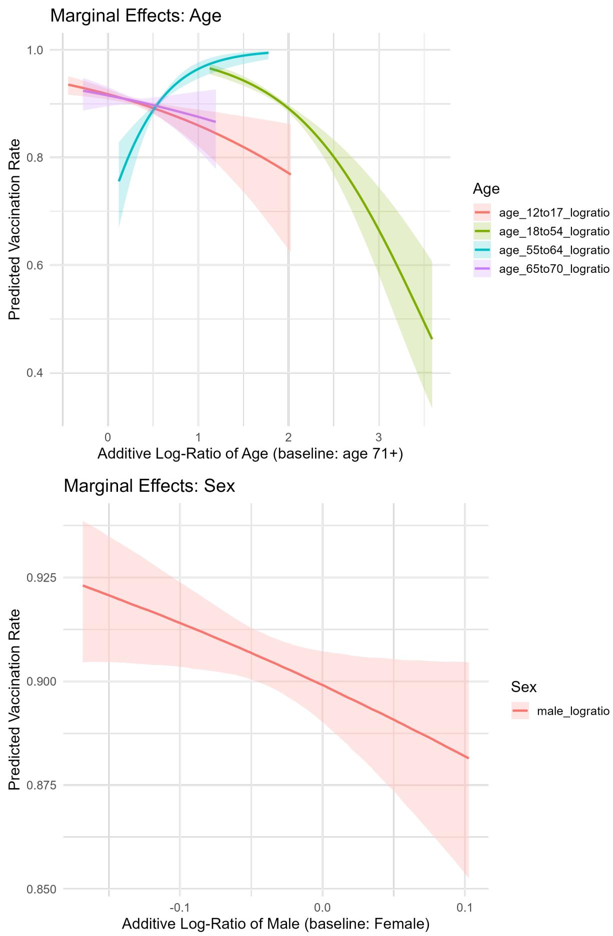

Figure 2: Base Model Marginal Effects of Adjusted Model

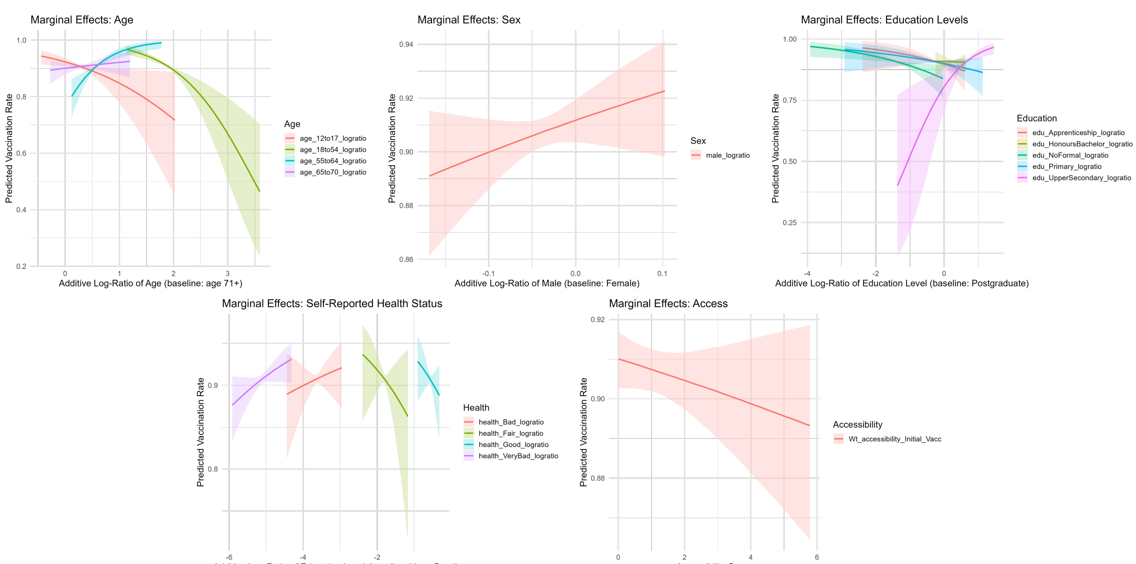

Figure 3: Marginal Effects of Adjusted Model

unexplained vaccination patterns) This confirmed that standard regression models failed to fully account for spatial dependencies.

To directly address spatial clustering, we implemented a CAR model, introducing a structured spatial random effect term that smooths values across neighboring areas. The spatial correlation parameter (car) was estimated at 0.56 (95% CI: 0.03 to 0.99), suggesting moderate positive spatial autocorrelation between neighboring LEAs, i.e., 56% of unexplained variance was attributable to spatial autocorrelation, indicating strong neighborhood-level similarities. This spatial clustering may reflect unmeasured factors such as local social networks, community attitudes, healthcare provider practices, or other geographically-clustered influences on vaccination behavior. The relatively modest spatial standard deviation (0.21) indicates that while spatial correlation exists, the magnitude of unexplained spatial variation is limited, suggesting that the measured covariates capture much of the systematic variation in vaccination rates. Demographic patterns remained robust: Age 55–64 still had the strongest positive effect on uptake. Upper secondary education, remains a large positive driver and healthcare proximity and healthcare outcomes continued to show near-zero effects. The consistency of these effects across spatial and non-spatial models suggests that educational disparities in vaccination uptake operate independently of geographic clustering and supply-side geographic barriers and self-reported health outcomes were not significant determinants of vaccination disparities.

The inclusion of the deprivation index had minimal impact on previously observed demographic and educational patterns. Age, education, health and accessibility effects remained consistent with prior models. The composite deprivation index showed a negligible association with vaccination rates (0.01 log-odds, 95% CI: -0.03 to 0.05), with the CI including zero. This finding suggests that area-level socioeconomic deprivation, as measured by the composite index, does not provide additional explanatory power beyond the specific educational attainment variables already included in the model. The minimal effect may indicate that educational composition captures the most relevant aspects of socioeconomic variation affecting vaccination behaviour, or that deprivation effects operate primarily through the measured demographic and educational pathways rather than through additional unmeasured mechanisms.

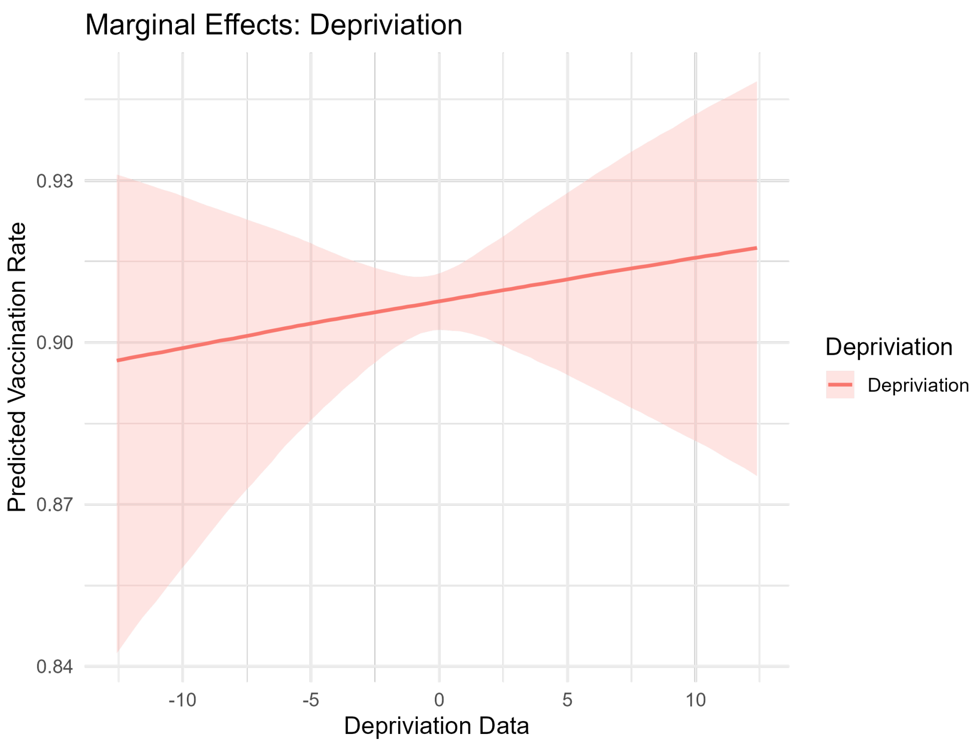

Figure 4: Marginal Effect of Depriviation

The age-sex interaction model revealed striking differences in age-vaccination relationships between males and females. Among females, younger age groups showed strong negative associations, particularly the 18-54 year group (-2.17 log-odds, 95% CI: -4.11 to -0.26), while the 12-17 year group showed a non-significant negative trend (-0.41 log-odds, 95% CI: -1.61 to 0.80). Older female groups demonstrated positive associations, with

55-64 year females showing 1.51 log-odds (95% CI: -0.17 to 3.23) and 65-70 year females showing 1.24 log-odds (95% CI: -0.29 to 2.76), though confidence intervals included zero for both groups. Younger male groups showed minimal negative effects, with 12-17 year males at -0.10 log-odds (95% CI: -1.16 to 1.01) and 18-54 year males showing a positive coefficient of 0.40 log-odds (95% CI: -1.23 to 2.08). The 55-64 year male group demonstrated a positive association (1.43 log-odds, 95% CI: -0.21 to 3.03) similar to their female counterparts. However, the 65-70 year male group uniquely showed a significant negative association (-1.86 log-odds, 95% CI: -3.34 to -0.34), contrasting sharply with the positive trend observed in older females. The strong negative association between middle-aged females (18-54) and vaccination contrasts with the near-neutral effect for middle-aged males, suggesting that vaccine hesitancy patterns differ substantially between genders within the same age cohorts. Conversely, the significant negative association for older males (65-70) versus positive trends for older females indicates that gender effects may reverse across age groups, with older men showing unexpectedly lower vaccination rates despite generally being considered a high-priority group.

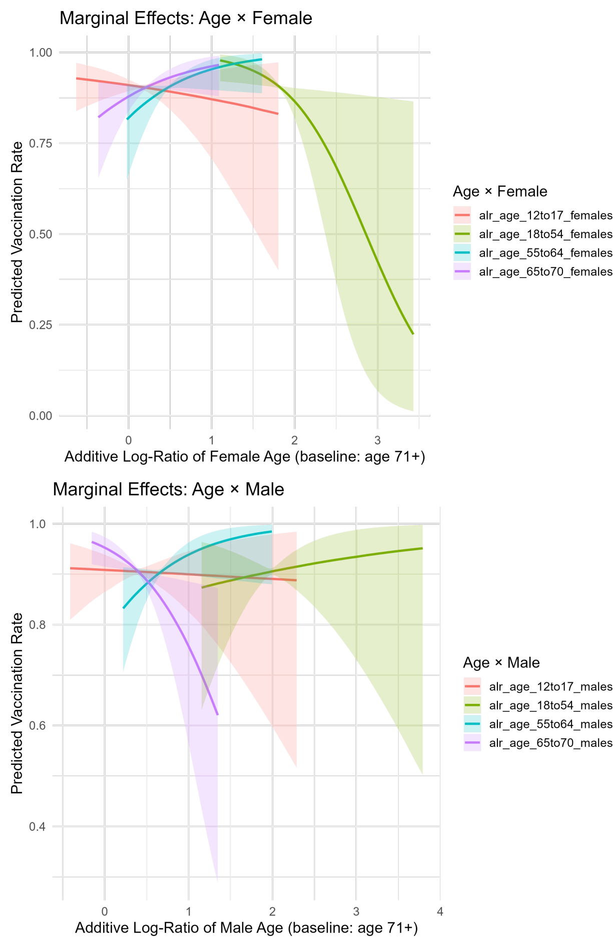

Figure 5: Marginal Effects of Age & Sex Interaction Model

We found distinct patterns in COVID-19 vaccination uptake trajectories across Ireland's LEAs using our fitted hierarchical beta regression model. The population-level upper asymptote was at 0.94 (95% CI: 0.76-1.12) on the logit scale, representing the baseline maximum vaccination rate before accounting for demographic influences. The temporal inflection point occurred at 5.61 months (95% CI: 5.57-5.66), indicating that the steepest increase in vaccination rates typically occurred during late May to early June 2021, corresponding to the period when vaccine availability expanded beyond priority groups. The scale parameter was found to be 1.36 (95% CI: 1.35-1.38), which controls the steepness of the sigmoidal curve transition and suggests a relatively gradual uptake pattern.

We identified substantial between-LEA heterogeneity in vaccination achievements, with the asymptote random effect standard deviation of 0.81 (95% CI: 0.73-0.91) indicating considerable variation in maximum vaccination rates between areas. The base asymptote showed more modest variation with a standard deviation of 0.25

(95% CI: 0.22-0.29), while the timing of peak uptake demonstrated relatively low variation between LEAs and the inflection point random effect standard deviation was only 0.26 (95% CI: 0.23-0.30), suggesting moderately synchronized temporal patterns across LEAs despite substantial differences in final vaccination levels.

The demographic predictors revealed positive associations between all covariates and final vaccination rate achievement. We found that 18-54 year olds showed a moderate positive effect on the asymptote (0.44 log-odds, 95% CI: 0.33-0.56), while 55-64 year olds demonstrated a stronger positive association (0.57 log-odds, 95% CI: 0.38-0.76). We observed that early retirement age populations (65-70 years) exhibited the strongest positive effect among age groups (0.64 log-odds, 95% CI: 0.45-0.82). Most notably, we found that male population proportion showed the largest positive association with ultimate vaccination rates (0.78 log-odds, 95% CI: 0.58-0.99), indicating that LEAs with higher male proportions achieved substantially higher maximum vaccination coverage.

The lower asymptote at -3.44 (95% CI: -3.49 to -3.39) on the logit scale, represents the minimal initial vaccination rates at the program's outset. Our demographic model maintained temporal stability, with the inflection point remaining consistent and the scale parameter showing minimal variation. We found that phi was 398.70 (95% CI: 382.35-415.69) which indicates low dispersion in the beta distribution, suggesting that our four-parameter logistic function effectively captured the systematic variation in vaccination trajectories while accounting for both demographic influences and spatial heterogeneity across Ireland's LEAs.

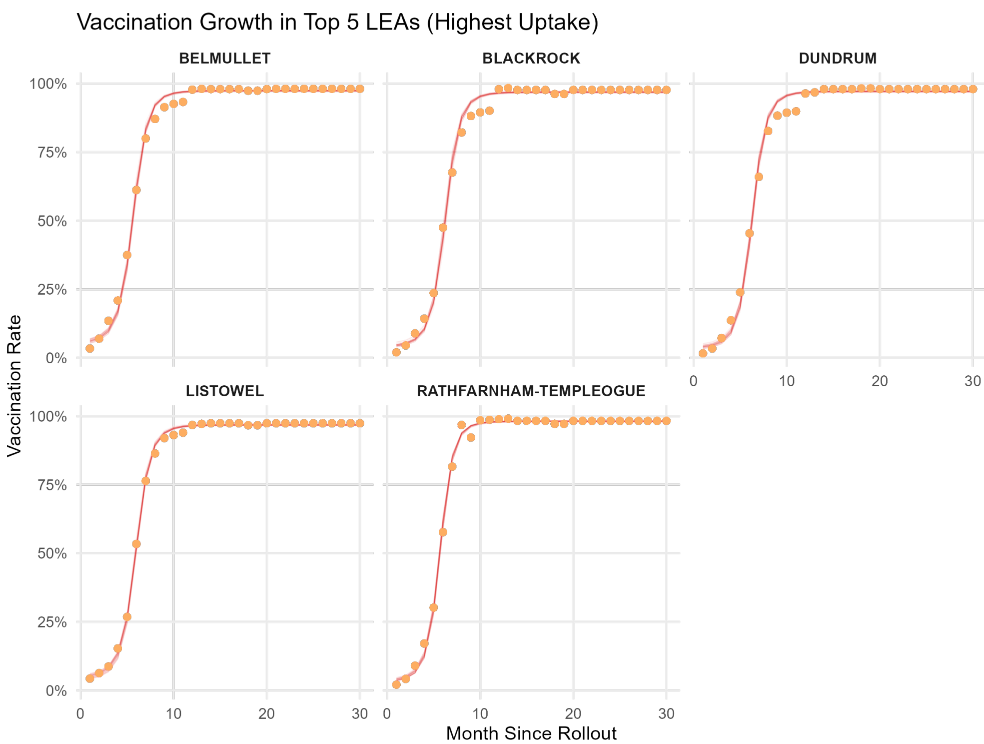

Figure 6: Vaccination Trend for Top 5 LEAs

4 Discussions

The adjusted base model demonstrated superior fit, as evidenced by the increased precision parameter (phi = 77.74 vs. 57.18), indicating reduced unexplained variance. The persistence of age effects across both models underscores the robustness of demographic patterns in vaccination behaviour, while the emergence of educational effects highlights the importance of socioeconomic factors. The negligible impact of initial vaccination center accessibility suggests that supply-side constraints were not primary drivers of vaccination disparities in this context, pointing instead to demand-side factors related to demographic and socioeconomic characteristics. The progressive increase in precision parameters across models (57.18 → 77.74 → 87.88) demonstrates continued improvement in model fit with the addition of socioeconomic variables and spatial

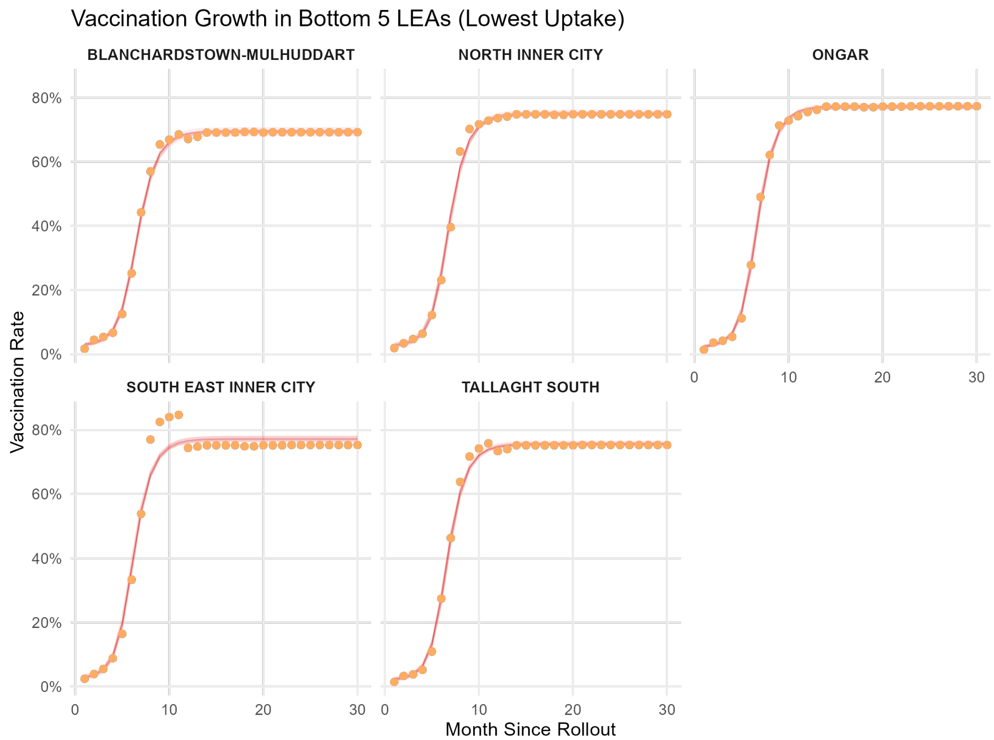

Figure 7: Vaccination Trend for Bottom 5 LEAs

structure and the spatial model represents the most comprehensive specification, accounting for both measured covariates and unmeasured spatially-correlated factors. The stability of key demographic and educational effects across all model specifications provides strong evidence for their robustness, while the significant spatial correlation highlights the importance of geographic context in understanding vaccination behaviour patterns. The longitudinal nonlinear model revealed nuanced demographic dynamics in vaccination uptake that differ from patterns observed in cross-sectional analyses. Specifically, the Asym was significantly and positively associated with the relative proportions of males and all working-age and older adult subgroups (ages 18–54, 55–64, and 65–70), with the strongest effects observed for the older cohorts. These findings suggest that, although younger adults and males may have exhibited delayed uptake in earlier phases of the campaign—as suggested in prior models—they ultimately reached high levels of vaccination over time. The estimated population-level midpoint for uptake (xmid aprox 5.61 months) indicates broad temporal alignment across LEAs, reflecting a relatively coordinated national rollout. However, substantial variation in asymptotic values between LEAs (sd(Asym) = 0.81) underscores the importance of local demographic composition in shaping final uptake levels. That the scalar parameter governing growth rate (scal = 1.36) was fixed across areas reinforces the interpretation that differences in eventual uptake—rather than uptake speed—were more influenced by population structure than by supply-side constraints. Together, these results indicate that extended temporal observation is critical for capturing the full extent of vaccination behaviour: initial disparities in uptake timing do not necessarily translate into persistent inequalities in ultimate coverage. Based on the observed vaccination rate data, the increase-then-decline patterns of vaccination rate identified in LEAs such as South East Inner City, Maynooth, Galway City Central, Tramore-Waterford City West, and Kenmare suggest anomalies when compared to the typical sigmoidal vaccination uptake curve. A likely explanation for this pattern lies in changes to the underlying population denominators used in rate calculations. Since vaccination rates are computed relative to estimated local populations, any mid-series updates—such as those informed by the 2022 Census, student migration, or seasonal population shifts—could result in apparent declines despite stable or increasing absolute vaccination numbers. This effect is especially plausible in areas with transient populations, such as university towns (Maynooth, Galway), city centres with high mobility (South East Inner City), and tourism-influenced or rural areas (Kenmare, Skibbereen-West Cork), where even small demographic changes can significantly distort percentage-based trends. As such, the observed dips are more likely to reflect population re-estimation or mobility rather than genuine decreases in vaccine uptake.

Our analysis relied on vaccination data reported only at the LEA level, not broken down by demographic subgroups or small area level data which was our biggest limitation. Population demographic characteristics were based on the 2022 Census and may not reflect the pandemic-period or vaccination roll out period scenario. Self-reported health status, used as a covariate, may be subject to bias. Our accessibility measure did not account for population demand or healthcare service's capacity, limiting its interpretive power; while a 2SFCA approach could address these spatial limitations, it requires more granular data and computational resources beyond this study's scope. Additionally, the analysis was restricted to the Republic of Ireland; cross-border vaccination—particularly in border regions—could have influenced local rates but remains unaccounted for in our models.

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